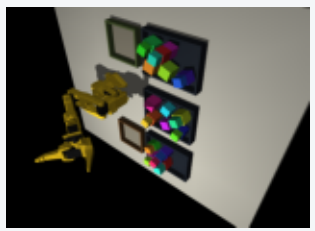
Front RGB + measured joint state $q_t$ (each control tick)

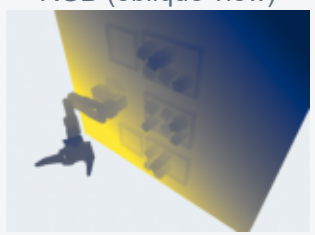
**Task specification**
Target attributes + destination / buffer

## Workspace

Two fixed RGB-D cameras



RGB (oblique view)



Metric depth (m)

**MuJoCo + SO-101**

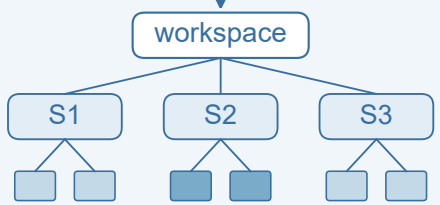
5 arm joints + gripper

## A Perception & HSG

**RGB masks + back-projection**
Color segmentation; calibrated depth

**3D fusion + box fitting**
Two views; configured box-size prior

**Temporal association**
Track IDs, geometry and history




Region hierarchy + current / held tracks

$G_t$

## B HSG planning

**Target grounding***
Attributes to an episode-local track

**Exact snapshot diff + indices**
Spatial hash and support relations

**Regional summaries**

| S1 | S2 | S3 |
|---|---|---|
| reuse | update | reuse |

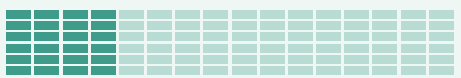
**Subgoal selection**
Unseen target: rank search regions
Visible target: resolve blockers
PICK / PLACE + placement pose

$g_k$

## C ACT control

**Observation / goal encoders**
Front RGB (CNN) + joint state
Subgoal to 25D numerical features

**Goal-conditioned ACT**
Transformer encoder-decoder



Predicted chunk: K = 16, 6 channels

**Chunk buffer + command limits**
4-step cache; one command per tick
Measured-state and joint-limit clipping

$u_t$ (30 Hz)

execute

refresh

replan

FK + tactile feedback

**CLOSED-LOOP ORCHESTRATOR**
Contract checks | execute | verify subgoal | re-observe and replan

Shown: RGB-D + indexed HSG + ACT. * Target grounding runs in the orchestrator.